

Chemistry Lab 5

DEGRADATION OF AN ORGANIC POLLUTANT IN WATER USING LED LIGHT AND A PHOTOCATALYST

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Abstract- This experiment deals with the study of the degradation of Methylene Blue (an organic dye, which is also regarded as a pollutant) in the presence of catalyst and light as well as without it to study the kinetics of the reactions. We will also use concepts such as Absorbance to determine the concentration of the pollutant.

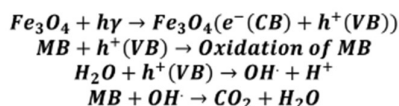
I. MATERIALS REQUIRED

Methylene Blue, Iron Oxide Nanoparticles (Fe_3O_4)

II. THEORY AND CONCEPTS

Methylene Blue is an Organic dye. It has a strong blue colour because of its absorbance spectrum. Using this absorbance spectrum, we can determine its concentration using its peak absorbance value at $\sim 664\text{nm}$. This value is given from the Beer Lambert's law, which states that Absorbance = $E \cdot c \cdot l$, where E is the molar extinction coefficient and it is a property of molecule. L was taken to be constant at 1 (in our experiment) and c is the concentration. We will use multiple readings to calibrate the value of E. Once we get the value of E, we can easily determine the concentration of solution at any stage of reaction.

The decomposition of Methylene Blue can be made faster using catalysts. One such catalysts is Iron Oxide nanoparticles. There is also light, which acts as a photocatalyst. The overall reaction is as given below:



Decomposition Reaction of Methylene Blue. Source: [1]

As we can see, the reaction is speeded up by the addition of the catalyst and the photocatalyst (light). The rate of the reaction can be quantitatively determined using the rate constant of the reaction, which can be interpreted from the value of $\log(C_0/C)$ at any stage of the reaction.

III. PROCEDURE

A. Part 1: Calibration with Known Concentration

In this part of the experiment, we will find the value of E. For this, we will take various quantities of Methylene Blue solution and correspondingly dilute them to find the absorbance at different values of concentration. We were given a 5ppm solution in the lab. The solution was divided continuously into different parts and diluted (as mentioned in the

results), and the corresponding absorbances were found, after which the mean slope between the concentration and absorbances was taken as the value of E.

B. Part 2: Study of Kinetics

5 ml of the solution was taken into three vials. In the first and the second vials, 5 mg of Fe_3O_4 nanoparticles were added as catalysts. The first vial was covered with aluminium foil. One magnetic bead was also added to vial 2. Vial 2 and vial 3 were then taken under an LED lamp, and vial 2 was continuously stirred using a magnetic stirrer. In every 10 minutes, the samples from the vials were taken, and their absorbances were measured. For vials 1 and 2, special care was taken by using strong magnets so that the Iron Oxide nanoparticles did not disturb the measurement process. The further calculations and plots are shown in the results section.

IV. RESULTS AND CALCULATIONS

A. Calibration

The test tubes T1, T2, T3, T4 and T5 had the following compositions

Test Tube	5ppm Methylene Blue	Water	Concentration
T1	2 ml	0 ml	5 ppm
T2	1 ml	1 ml	2.5 ppm
T3	1 ml	2 ml	1.67 ppm
T4	1 ml	3 ml	1.25 ppm
T5	1 ml	4 ml	1 ppm

Table 1. Concentrations in various Test Tubes

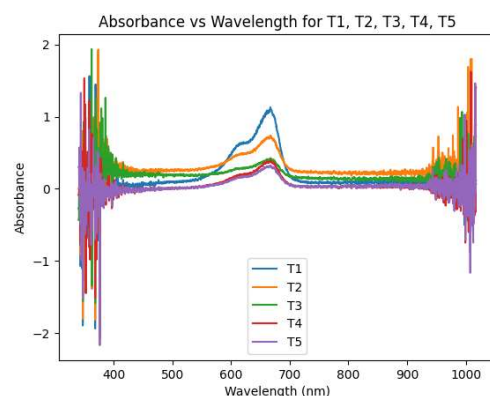


Fig. 1. Plot of Absorbance against Wavelength for different test tubes.

Test Tube	Peak Value (Absorbance)
T1	1.1301
T2	0.7367
T3	0.4251
T4	0.3967
T5	0.3247

Table 2. Peak absorbance values at ~664nm.

Performing the calculation for E, that is, the average slope between Absorbance and Concentration, we get the plot as follows.

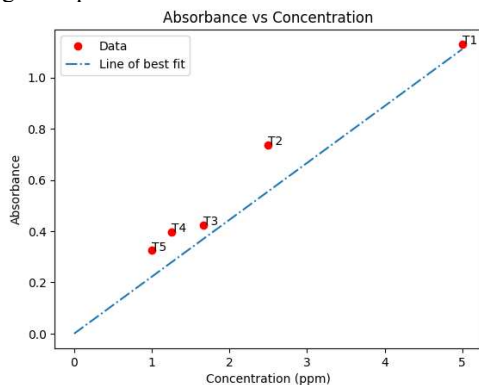


Fig. 2. Plot between the absorbance values against known concentrations

The value of E was found to be 0.22 Units.

B. Vials

The absorbance for the vials at Time 0 was found to be 0.6867 (Absorbance units). Therefore, using Beer Lambert's law, we get the concentration of the unknown solution to be 3.09 ppm.

The plots for the absorbance values for all the vials are as follows.

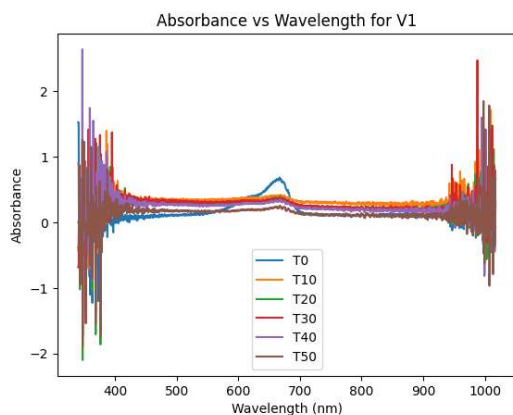


Fig. 3. Absorbance Spectrum in V1 at different times

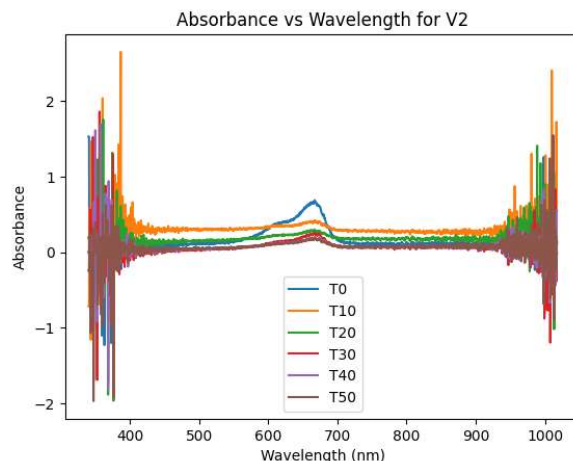


Fig. 4. Absorbance Spectrum in V2 at different times

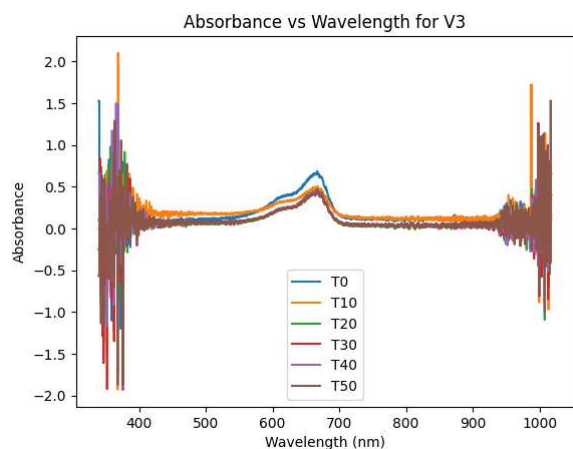


Fig. 5. Absorbance Spectrum in V3 at different times

Based on these plots, the values of the corresponding peak values were found. These peak values of absorbances were then used to find the concentrations at different times using the previously calculated value of E.

Vial	Conc. at 20 mins	Conc. at 50 mins
V1	1.74 ppm	1.51 ppm
V2	1.35 ppm	0.85 ppm
V3	2.12 ppm	2.02 ppm

Table 3. The values of concentration at times 20 minutes and 50 minutes

We can clearly see that the amount of decrease in concentration was the most in vial 2, which had both catalyst and light. This was followed by vials 1 and 3, which only had catalyst or light, respectively. These values of concentrations at different times were then used to plot the curves of $\log(C_0/C)$, which roughly gives us the rate of the reaction.

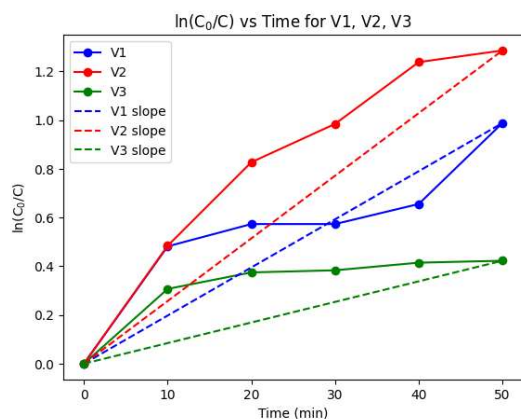


Fig. 6. Log (C_0/C) vs t for vials 1,2,3.

The slopes for different Vials are as follows:

Slope for V1 is 0.019757852666589793

Slope for V2 is 0.025718532589016064

Slope for V3 is 0.008452998657245305

C. Observations

The reaction speeds up on the addition of both the catalyst and the light (photocatalyst). That is why, the rate of reaction is the fastest in the V2. The reaction was the slowest in vial 3, which lacked the catalyst. The reaction was moderately fast in vial 1, which contained the catalyst but was covered with light. So, we can observe that the reaction is the fastest in vial 2, followed by vial 1 and vial 3. This is also reflected in the slope values of the curves as provided in the results section. However, there are some inaccuracies in our measurement. We cannot expect a sudden rise in the reaction rate between T40 and T50 for vial 1. This deviation from the ideal rate is most probably because of the exposure to light that happened during the measurement process near the end of the experiment. The ideal expectation is that the curves will start saturating as we move ahead in time, but that was not shown in the case of V1 due to the above-mentioned reason. Though the measurement values may not be very accurate, the entire aim of the experiment to understand the speed up of the reaction due to catalysts and photocatalysts was also fulfilled by practically observing the rate at which the strong blue colour of the solution was fading away.

V. REFERENCES

[1] Lab Manual

VI. MEMBER CONTRIBUTIONS

Jaidev	Experimentation, Report Development
Pranav	Experimentation, Report Review
Sibtain	Experimentation, Report Review
Suteekshna	Experimentation, Report Review